

Insurance Portfolio Intelligence

Management report and technical review

This report examines the supplied insurance portfolio project, its stored Power BI data and its reporting logic. It explains portfolio composition, modeled premium exposure and the controls needed before the dashboard supports operational decisions.

Executive assessment

The active portfolio contains 7,332 records and \$25.43 million in modeled annual premium. Minnesota, Wisconsin and New Jersey account for 33.71% of that premium. These concentrations identify the largest geographic segments within the project dataset and support a focused portfolio review.

Active portfolio indicator	Verified value
Record count	7,332
Modeled annual premium	\$25,430,623.29
Modeled premium paid	\$125,721,742.50
Modeled remaining lifetime premium	\$334,238,049.12

The main management priority is data reliability. Region is missing for 76.30% of all records, while maturity and return calculations require a business-rule review. Those measures should not guide financial decisions until their assumptions are validated.

Value for management and HR

For management, the project makes portfolio mix and premium assumptions visible. For HR and hiring reviewers, it provides concrete evidence of Power Query preparation, dimensional modeling, DAX, report design and analytical quality review. It does not demonstrate measured revenue improvement, campaign effectiveness or employee performance.

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Reporting conventions

Active scope means Policy Status equals Active, matching the stored report-level filter. All-status scope includes Active, Surrendered, Lapsed and Claimed records. Each analytical section states its scope.

Records are fact-table rows, not necessarily unique policies. The dataset contains 10,000 rows but 9,938 distinct policy numbers. Customer counts are separate and must not be substituted for policy counts.

Report monetary findings use USD. M means millions. Screenshot totals reflect the filter selections shown on each page.

“Modeled paid” and “modeled remaining” follow the project’s annual-premium and duration formulas. They are not verified cash collections, overdue balances or forecasts of recoverable revenue.

State findings describe this portfolio; wider market representativeness has not been established.

Project scope and data

The project brings policy records together with customer, agent, product and management dimensions. Its business purpose is to make portfolio composition, premium measures and the sales hierarchy accessible within one Power BI report.

Source table	Rows	Role
Insurance policy fact	10,000	Policy records and premium inputs
Customer detail	9,945	Customer attributes
Insurance agent	22	Agent identifiers and contact fields
Protection plan	8	Product names and plan attributes
Policy type	3	Product classification
Regional manager	5	Regional assignments
Zonal manager	3	Management hierarchy

Coverage and model structure

The fact data covers 22 U.S. states. Purchase dates run from July 24, 2015 to July 23, 2025. The model contains seven CSV-backed queries and two derived queries for the hierarchy and region list. Eight business relationships connect the fact and supporting dimensions; automatic date structures are outside that count.

The hierarchy has 23 state entries, including Missouri, which has no fact records. This is partial geographic coverage, not a nationwide sample. No sampling method or evidence of population representativeness was provided.

Source preparation

The source files contain policy records, U.S. state values and USD monetary amounts. Power Query removes preamble rows, promotes the expected headers and prepares the fields needed by the model. Customer, agent, product and management identifiers connect descriptive attributes to the policy records.

Coverage, premium, loan allowance and underwriting amounts provide distinct analytical inputs. Annual and lifetime premium measures are then calculated from payment frequency and tenure. Clear field definitions help prevent these different quantities from being interpreted as interchangeable.

Scope boundary

The materials contain insurance portfolio data, not advertising performance data. No campaign spend, impressions, clicks or acquisition metrics were supplied. Marketing attribution, return on advertising spend and campaign comparisons are therefore outside this report.

Portfolio composition

Scope: all 10,000 records. The active category represents 73.32% of records; surrendered records account for 25.24%. These are status shares in a stored snapshot, not annual retention or churn rates.

Stored policy status	Records	Share
Active	7,332	73.32%
Surrendered	2,524	25.24%
Lapsed	107	1.07%
Claimed	37	0.37%

Product concentration

Protection plan	Records	Share
ULIP Growth Plan	3,205	32.05%
Life Assurance	2,358	23.58%
Life Growth Advantage	1,818	18.18%
Smart Term Protect	1,236	12.36%
Income Protection Plan	972	9.72%
Whole Life Shield	329	3.29%
Retire Secure	65	0.65%
Guaranteed Wealth Builder	17	0.17%

ULIP Growth Plan and Life Assurance together represent 55.63% of all records. This concentration suggests that product-level data and service reviews should start with these two plans. Record share alone does not establish product profitability or suitability.

Patterns requiring investigation

All 107 lapsed records and all 37 claimed records are assigned to Indiana. Validate the source segmentation before interpreting this as a location effect. The data does not identify a cause for either status.

Quarterly, monthly and annual payment frequencies contain 3,432, 3,324 and 3,244 records respectively. Frequency normalization is essential when comparing premium amounts across these groups.

Premium exposure and concentration

Scope: 7,332 active records, covering 7,299 distinct customer IDs. Annual premium is normalized by payment frequency; lifetime amounts also depend on the recorded tenure.

Modeled premium measure	USD	Share of lifetime total
Lifetime premium	\$459,959,791.62	100.00%
Premium paid	\$125,721,742.50	27.33%
Remaining premium	\$334,238,049.12	72.67%

The remaining amount is the difference between modeled lifetime premium and modeled paid premium. It includes future tenure and is not an arrears measure. Without a payment ledger, it cannot be treated as a collections target.

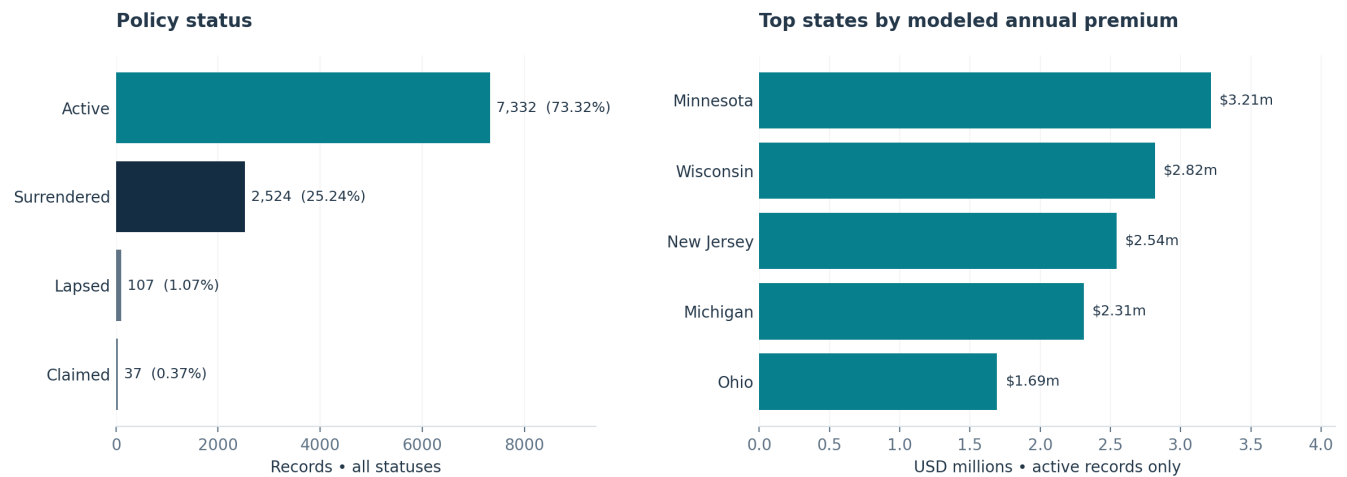


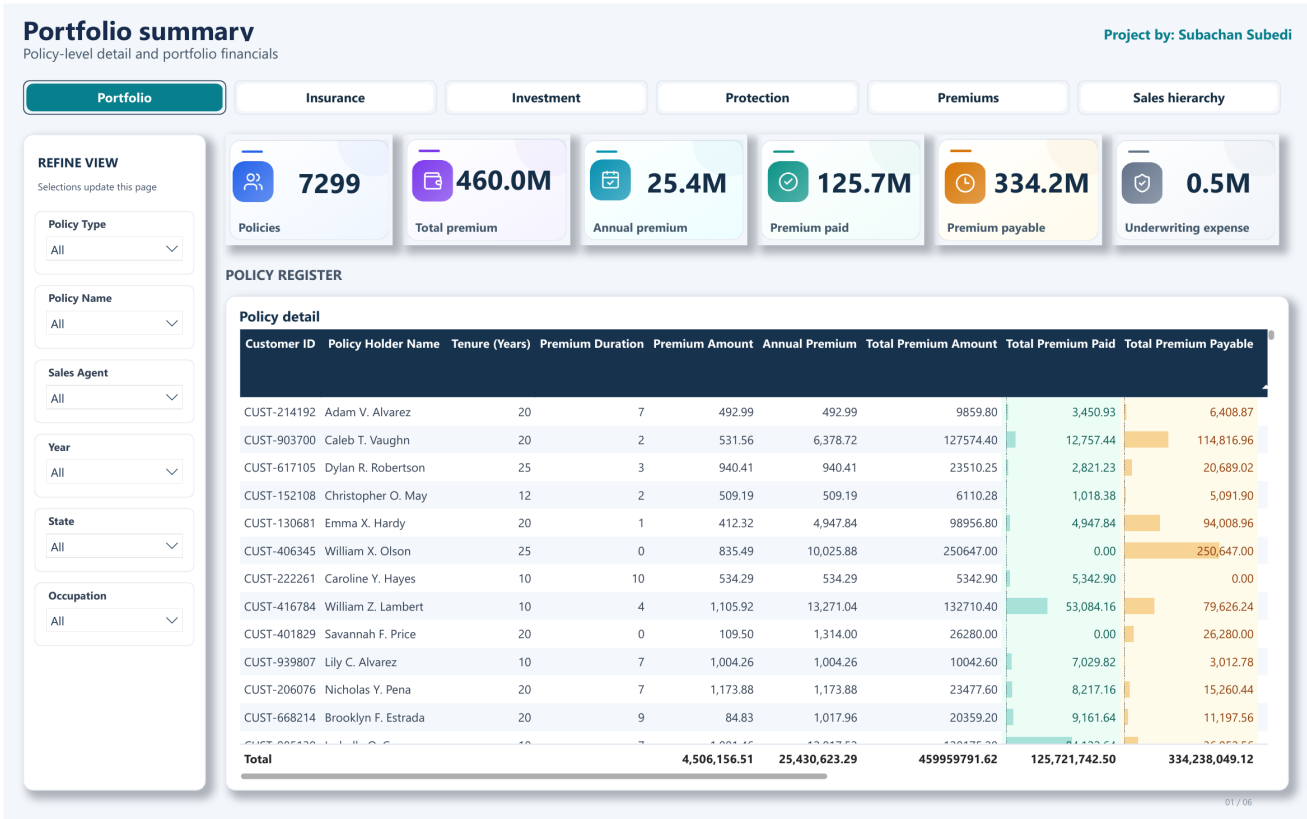
Figure 1. Recomputed from stored model data. Left: all statuses. Right: active records. Amounts are USD. This is an analytical summary, not a dashboard screenshot.

Concentration that matters

Minnesota contributes \$3.21 million, Wisconsin \$2.82 million and New Jersey \$2.54 million in active modeled annual premium. Together, their unrounded values represent 33.71% of the active total. This highlights concentration within the portfolio; it does not establish demand or growth potential in the wider market.

The model also contains \$90.86 million in active coverage amounts. Coverage and premium describe different quantities and should remain separate in cards, tooltips and management discussions.

Portfolio summary



Dashboard 1. Source: insuranceusbased.pdf, page 1. Screenshot reproduced from the supplied export.

What this page shows

The starting view combines headline portfolio measures with the policy register. The visible slicers cover policy type, protection plan, sales agent, year, state and occupation.

Reading the captured view

The export shows 460.0M total premium, 25.4M annual premium, 125.7M modeled paid premium and 334.2M modeled remaining premium. These rounded cards agree with the active-scope totals in this report. The 7,299 card is labeled Policies in the export; the inspected model identifies this as the active customer count, not the count of fact records.

Management use

Use the register to trace totals to customer-level rows and compare tenure, payment duration and premium values. Conditional data bars help distinguish modeled paid and remaining amounts. The screenshot shows only the visible part of the scrollable table.

Insurance overview



Dashboard 2. Source: insuranceusbased.pdf, page 2. Screenshot reproduced from the supplied export.

What this page shows

This page breaks annual premium down by policy type, protection plan, agent, state, occupation and policy start year. The measure selector is set to Total Annual Premium in the supplied export.

Reading the captured view

The state chart places Minnesota, Wisconsin and New Jersey first. The product chart shows ULIP Growth Plan at 32.74% and Life Assurance at 23.34% of active annual premium. These premium shares answer a different question from the all-status record shares reported earlier.

Management use

Use the page to identify concentration, compare product contributions and select segments for follow-up. The yearly chart describes the selected portfolio by start-year cohort; it should not be interpreted as a cash-receipt history or a year-over-year revenue growth measure.

Investment and maturity



Dashboard 3. Source: insuranceusbased.pdf, page 3. Screenshot reproduced from the supplied export.

What this page shows

The combined chart compares total lifetime premium and modeled maturity by policy start year. A separate percentage axis displays Average of Annualized ROI.

Reading the captured view

The screenshot retains the portfolio summary cards while showing the financial comparison below. Its two column series use monetary amounts; the line uses a percentage. Reading both against one scale would give the wrong interpretation.

Management use

Use this page to examine the assumptions linking lifetime premium and projected maturity. The displayed maturity and ROI series remain subject to the calculation review described in the limitations section. Their presence in the screenshot does not establish realized investment performance.

Premium and protection



Dashboard 4. Source: insuranceusbased.pdf, page 4. Screenshot reproduced from the supplied export.

What this page shows

This page compares annual premium, coverage amount and Average of Annualized ROI by tenure year. The supplied screenshot applies a Tenure Years range of 20 to 25.

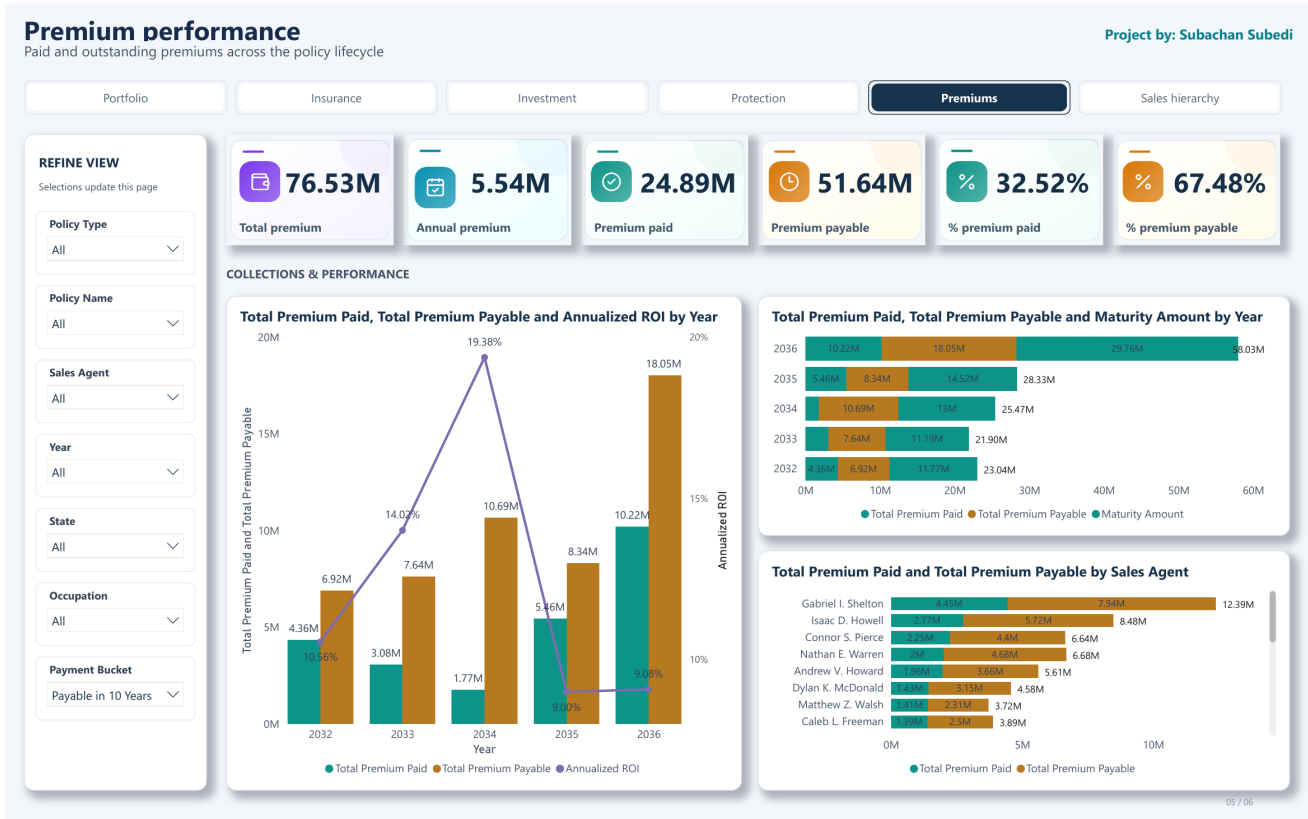
Reading the captured view

The filtered cards show 358.7M total premium, 16.4M annual premium, 81.8M modeled paid premium and 276.9M modeled remaining premium. These are the values for the visible selection and should not be compared directly with the unfiltered active portfolio as though the populations were identical.

Management use

Use the chart to compare premium contributions with the protection recorded for the selected policies. Coverage is separate from premium receipts. The line uses the right-hand percentage axis and remains subject to validation of the return calculation.

Premium performance



Dashboard 5. Source: insuranceusbased.pdf, page 5. Screenshot reproduced from the supplied export.

What this page shows

The export selects Payment Bucket equal to Payable in 10 Years. It shows modeled paid and remaining premium by tenure year, a comparison including maturity, and an agent-level premium breakdown.

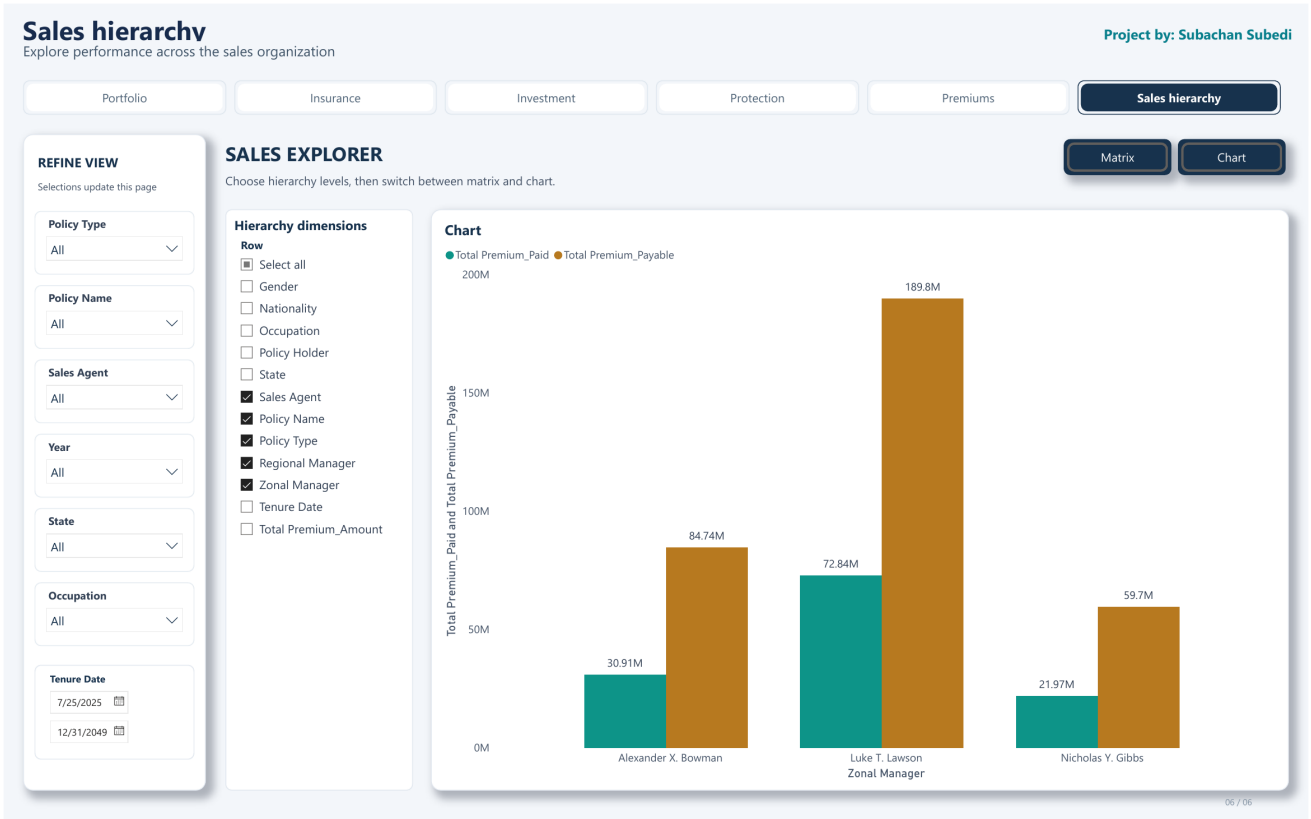
Reading the captured view

For this selection, the cards display 76.53M total premium, 5.54M annual premium, 24.89M modeled paid premium and 51.64M modeled remaining premium. The displayed paid and remaining shares are 32.52% and 67.48%. These percentages apply to this filtered view, rather than the full active portfolio.

Management use

Use the agent chart and year groups to identify where reconciliation should begin. Remaining premium includes future tenure and is not an overdue balance. In the upper-right stacked chart, paid, remaining and maturity are different concepts; its combined bar length should not be treated as one financial KPI.

Sales hierarchy



Dashboard 6. Source: insuranceusbased.pdf, page 6. Screenshot reproduced from the supplied export.

What this page shows

This screenshot shows the Chart view grouped by zonal manager. The hierarchy selector includes agent, protection plan, policy type, regional manager and zonal manager selections; Matrix and Chart buttons are visible.

Reading the captured view

The date range shown is July 25, 2025 to December 31, 2049. Among the three displayed groups, Luke T. Lawson has the largest modeled paid and remaining premium amounts: 72.84M and 189.8M. These rounded values describe the captured view, not an employee-performance score.

Management use

Use the hierarchy to trace premium amounts through organizational groupings and compare totals under the same filters. The PDF captures the chart state only; the matrix is an alternative available in the report definitions. Validate regional assignments and access rules before relying on manager-level reporting.

Technical approach and KPI definitions

Power Query and the semantic model

CSV queries remove source preambles, promote headers and apply the expected schema. The fact query joins a derived state hierarchy to obtain Region. The hierarchy splits regional state lists into rows, and the Region query produces the distinct region list.

Six fact-to-dimension relationships connect customer, agent, zonal manager, policy type, regional manager and protection plan identifiers. The remaining two business relationships connect the hierarchy to regional managers and the fact Region field to the Region table. Seven use bidirectional filtering and one uses single-direction filtering.

All six primary dimension keys are unique, with no unmatched non-null foreign keys in the stored fact table. This supports identifier integrity, but does not resolve the missing Region values introduced by the separate state-based join.

How the headline values are calculated

Metric	Implementation in the inspected model
Annual premium	Source premium × frequency factor: annual 1, quarterly 4, monthly 12; half-yearly 2 is supported by the formula.
Lifetime premium	Annual premium × recorded tenure.
Payment duration	DATEDIFF between Start Date and Last Paid Date, using YEAR. This counts year boundaries.
Modeled premium paid	Annual premium × payment duration.
Modeled remaining premium	Lifetime premium minus modeled premium paid.
Paid share	Sum of modeled paid premium ÷ sum of lifetime premium.

The model contains 11 explicit measures alongside calculated columns. The headline premium totals above were independently aggregated from the extracted model. Percentages use aggregated amounts, rather than averaging row-level percentages.

Filter context and reconciliation

The active report filter is applied before aggregating headline premium measures. Record counts, customer counts and distinct policy counts answer different questions. Reconcile a sample of filtered records to the underlying calculations before relying on a new visual or comparison.

Data quality and limitations

Regional assignment needs repair

Region is blank for 7,630 of 10,000 records, or 76.30%. Among active records, 5,576 are blank. The hierarchy query splits comma-separated state lists without trimming the resulting tokens; leading spaces prevent many exact state matches. Trim both sides of the join and reconcile coverage before using Region-based comparisons.

Record grain requires a business decision

There are 9,938 distinct policy numbers across 10,000 fact rows: 62 rows beyond the distinct-ID count. This does not by itself prove duplicate records. Confirm whether repeated identifiers represent valid events or an unintended grain before reporting a unique-policy KPI.

Return measures are not decision ready

The maturity calculation contains premium thresholds, tenure-based rates and product-specific conditions that require business validation. One product-name condition does not match the current product dimension. Maturity is zeroed for non-active records, which changes the interpretation of the derived profit measure. These rules need reconciliation before use.

The calculated Annualized ROI field is blank for 780 records. The CAGR measure combines aggregated maturity and modeled paid premium with the maximum tenure; it is not a transaction-based investment return. No maturity, profit or ROI headline is endorsed in this report.

Access controls need role testing

Agent access is defined through an email comparison to USERPRINCIPALNAME. The regional-manager role compares a region field with the signed-in email instead of the looked-up region. The zonal-manager role returns an owner branch or BLANK without an explicit assigned-row condition. These expressions need correction and role-specific tests; their presence is not evidence of effective access control.

Operational and publication boundaries

Queries still contain local source paths. A clean refresh needs an accessible data location and compatible headers. The supplied PDF confirms the exported appearance of all six pages. It does not verify service deployment, scheduled refresh, live data quality or interactive filter behavior.

The dataset has not been established as a representative U.S. market sample. No verified cash ledger, claim-payment amounts, employee outcomes, campaign results or measured business uplift were supplied. These gaps prevent conclusions about realized collections, profitability, HR performance or marketing effectiveness.

Recommendations and evidence register

Recommended sequence

1. Restore analytical trust. Trim state tokens, rebuild the regional join and reconcile missing assignments. Confirm the fact-table grain and use explicit row-count, distinct-policy and customer-count labels. Proposed completion evidence: reconciled coverage and documented grain rules.
2. Reconcile financial definitions. Validate premium thresholds, tenure-based rates and product-name conditions. Keep modeled premium measures separate from actual collections. Review return logic with the business owner before displaying it as a financial KPI.
3. Verify the user experience and access. Refresh from the intended data location and test page navigation, slicers, totals and each security role in Power BI Desktop. Confirm that page titles, filter context and financial labels remain clear in the final dashboard export.
4. Investigate the portfolio patterns. Review the two largest plans, the top three premium segments and the concentration of lapsed and claimed records. Treat these as questions for follow-up, not proof of causes or grounds for immediate commercial action.

Outcomes supported by the evidence

The project brings seven source tables into a connected reporting model, standardizes premium comparisons and provides six analytical pages. This review adds verified totals, scope-aware findings and a prioritized list of model issues. The materials do not quantify time saved, revenue gained or operational improvement.